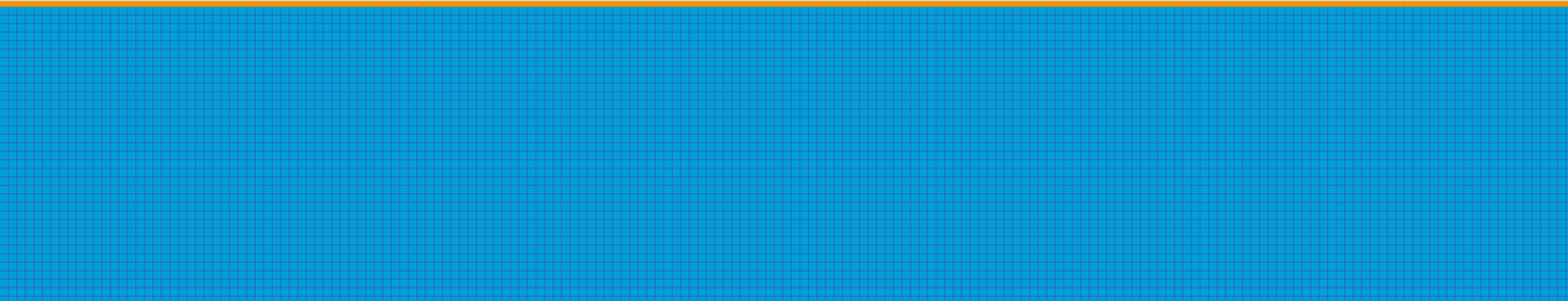
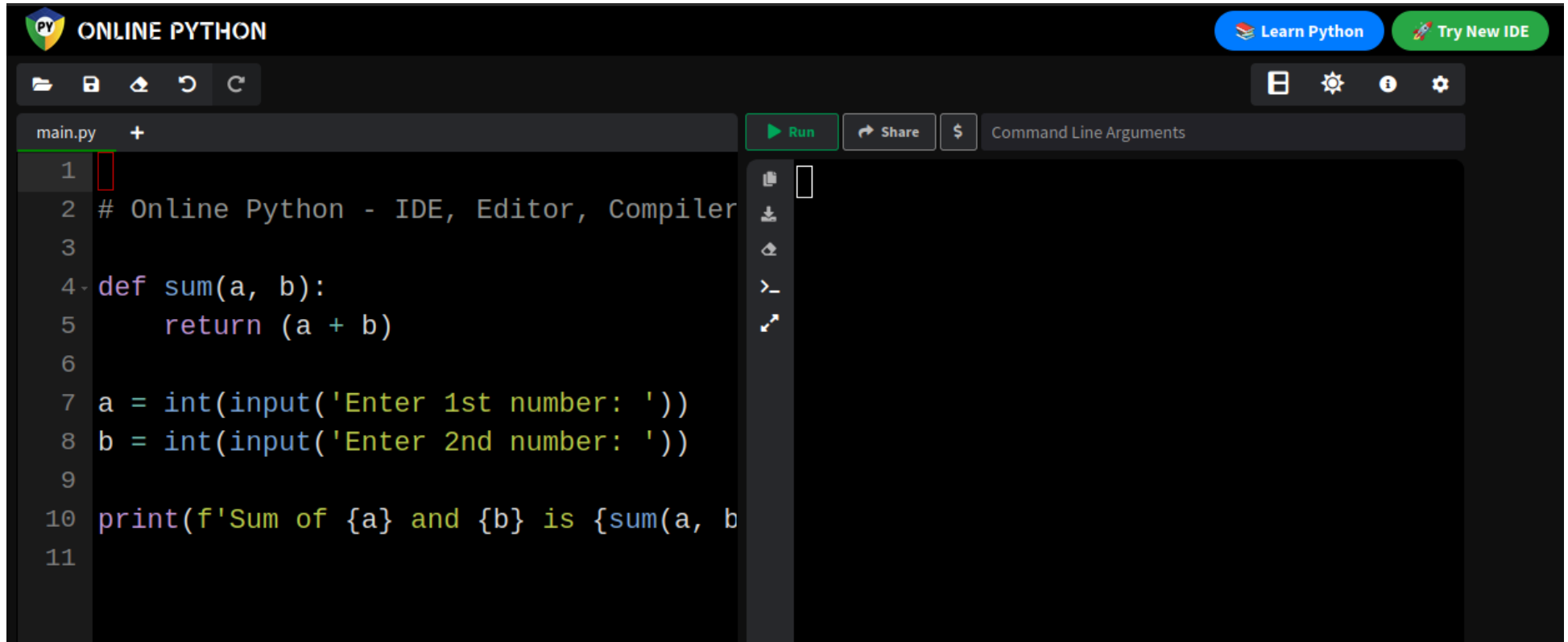


Introduction to Python



Get Coding

<https://www.online-python.com/>



The screenshot displays the Online Python IDE interface. At the top, there's a header with the 'ONLINE PYTHON' logo and two buttons: 'Learn Python' and 'Try New IDE'. Below the header is a toolbar with icons for file operations (new, open, save, undo, redo) and settings. The main editor area shows a file named 'main.py' with a Python script. The script defines a function 'sum(a, b)' that returns 'a + b', and then uses it to calculate the sum of two input numbers. The output area on the right is empty.

```
1  
2 # Online Python - IDE, Editor, Compiler  
3  
4 def sum(a, b):  
5     return (a + b)  
6  
7 a = int(input('Enter 1st number: '))  
8 b = int(input('Enter 2nd number: '))  
9  
10 print(f'Sum of {a} and {b} is {sum(a, b)}')  
11
```

Getting Started

Hello World:

The simplest program, this uses the *print* function to display *Hello, World*

Code:

```
print("Hello, World")
```

Output:

```
Hello, World
```

Variables

Variables are a place to store data values, variables are created by assigning a value to it.

example:

```
x = 5  
y = "Billy"
```

```
x = 5  
y = "Billy"  
  
print(x)  
print(y)  
  
  

```

```
5  
Billy
```

Data types

In programming data types are important. Variables can store different data types and can do different things with different data types.

example(These are not all the data types):

Type	Name	Example
Text	str (string)	"Billy"
Number	int (integer), float, compls	int: 1 float: 3.14
Boolean	bool	1/0 True/False

Numbers

Numbers can be whole or decimal or complex. We will focus on the first two.

example:

$x = 1 \leftarrow$ this is an int or an integer

$y = 3.1124125 \leftarrow$ This is a float or floating point number

Operators

Operations such as Addition(+), Subtraction(-), Multiplication(*), Division(/) Are the basic Arithmetic Operators.

Code:

```
print(5 + 4)
```

Output:

```
9
```

Logical Operators

There are other operators like comparison operators or logical operators.

Logical operators are `and`, `or` and `not`.

Say, $x = 5$

Operator	Name	Example	Result
and	TRUE if both statements are TRUE	$x < 6$ and $x < 10$	TRUE
or	TRUE if one Statement is true	$x < 7$ or $x < 4$	TRUE
not	FALSE if the result is TRUE	not($x < 5$ and $x < 10$)	FALSE

Comparison Operators

Comparison operators: $x = 5$, $y = 5$

Operator	Name	Example	Result
==	Equal	$x == y$	TRUE
!=	Not Equal	$x != y$	FALSE
>	Greater than	$x > y$	FALSE
<	Less than	$x < y$	FALSE
>=	Greater than or Equal to	$x >= y$	TRUE
<=	Less than or Equal to	$x <= y$	TRUE

IF...

Using the Logical and Comparison Operators we can check `IF` statements are true or false.

Code:

```
x = 1000↵  
y = 100↵  
if x > y:↵  
    print("x is greater than y")↵
```

Output:

```
x is greater than y
```

ELSE

Code:

```
x = 10
y = 100
if x > y:
    print("x is greater than y")
else:
    print("y is greater than x")
```

Output:

```
y is greater than x
```

ELIF

elif is when the previous if statement returns false you can add another if statement to check multiple expressions for true.

Code:

```
x = 10
y = 10
if x > y:
    print("x is greater than y")
elif x == y:
    print("x is equal to y")
else:
    print("y is greater than x")
```

Output:

```
x is equal to y
```

Loops

There are two main types of loops, `For loops` and `While loops`

While loop:

```
i = 1
while i < 10:
    print(i)
    i += 1
```

For loop:

```
fruits = ["apples", "bananna", "pear"]
for fruit in fruits:
    print(fruit)
```